

EECE7352 Computer Architecture

Homework 2

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Part D:

1) Name 2 CISC instruction set architectures, and 2 RISC instruction set architectures.

Ans:

CISC: VAX and Intel 8086

RISC: ARM and MIPS

2) Describe 3 characteristics of the DEC Alpha instruction set.

- It is a true **64-bit architecture** (strictly 64 bit)
- It used a pure **load/store** Architecture
- **32-bit** fixed instruction length
- **32** General Purpose Registers and 32 Floating Point Register, each 64 bit wide.

3) Discuss the differences/similarities between MIPS and PowerPC in terms of how they handle conditional branches.

Similarities:

- Both MIPS and PowerPC are RISC Architectures.
- Both Arch have dedicated instructions designed to alter the flow of program execution based on certain conditions.

Differences:

- For Condition Evaluation, in MIPS the branch instruction itself performs the comparison whereas in PowerPC these are two separate steps. A cmp instruction first compares values and sets flags in the dedicated Condition Register (CR).
- For state management there are no persistent flags in MIPS whereas there is a dedicated Condition Register to store the results of comparisons in PowerPC

4) Provide an example of how register windows work on the SPARC ISA.

SPARC introduced register windows back in 1987, where the register file is divided into overlapping windows that shift on procedure calls.

Advantage:

Reduces the need for saving registers to memory across procedure calls, speeding up function execution.

Disadvantages:

Requires extra hardware registers and later became problematic for superscalar implementations.

5) In your opinion, which generation of the Intel x86 architecture provided the most significant advances as compared to the previous generations of the ISA.

After going through the Introductory history of Intel Architecture in the Appendix, the Intel 80836 launched in 1985 extended the 16-bit 80286 architecture to 32 bits. This included 32-bit registers and a 32-bit address space, dramatically increasing the amount of memory the processor could directly access and the complexity of the data it could handle in a single operation.

80836 also introduced Paging support, a crucial feature for modern multitasking operating systems like Windows, macOS, and Linux, which was an addition to the segmentation model from the 80286.